

Lezione del 28-5-2026

giovedì 28 maggio 2026 09:09

De limite notevole

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

(F.L. $\frac{0}{0}$)

$$\frac{1 - \cos x}{x^2} = \frac{1 - \cos x}{x^2} \cdot \frac{1 + \cos x}{1 + \cos x} = \frac{1 - \cos^2 x}{x^2 (1 + \cos x)} \quad \begin{matrix} \text{Formula} \\ \text{Fondamentale} \end{matrix}$$

$$= \frac{\sin^2 x}{x^2 (1 + \cos x)} = \left(\frac{\sin x}{x} \right)^2 \cdot \frac{1}{1 + \cos x}$$

10. (continued)

$\backslash \quad \cdot \quad \cup \quad \text{Case } k+2$

$x \rightarrow 0$
 $(1)^2$
 $\frac{1}{2}$
 $\frac{1}{2}$

Pertanto

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

Let $\lim_{x \rightarrow x_0} f(x) = 0$, where

$$\lim_{x \rightarrow x_0} \frac{1 - \cos(f(x))}{[f(x)]^2} = \frac{1}{2}$$

Ex

$$\lim_{x \rightarrow 1} \frac{1 - \cos(\sqrt{x-1})}{x-1}$$

$$\begin{pmatrix} 0 & F \end{pmatrix}$$

$n \rightarrow \infty$

$$e^{n-1} - 1$$

$$\left(\frac{0}{0} \right)$$

$$\frac{1 - \cos(\sqrt{|n-1|})}{e^{n-1} - 1} = \boxed{\frac{1 - \cos \sqrt{|n-1|}}{(\sqrt{|n-1|})^2}} \cdot \frac{(\sqrt{|n-1|})^2}{e^{n-1} - 1}$$

A parte colapso

$$\lim_{n \rightarrow \infty} \frac{|n-1|}{e^{n-1} - 1}$$

$$\left(\frac{0}{0} \right)$$

$$\left(\frac{0}{0} \right) \quad e^{f(n)} - 1$$

$$\left(\begin{array}{c} \text{num} \\ n \rightarrow n^* \end{array} \right) \frac{\quad}{\text{den}} \quad -1$$

$$\frac{|n-1|}{e^{n-1} - 1} = \frac{|n-1|}{e^{n-1} - 1} \cdot (n-1) = \frac{|n-1|}{n-1}$$

$$= \left(\frac{1}{e^{n-1} - 1} \right) \cdot \left(\frac{|n-1|}{n-1} \right)$$

$n \rightarrow 1^+ \rightarrow 1$
 $n \rightarrow 1^- \rightarrow -1$

$n \rightarrow 1 \rightarrow \frac{1}{1} = 1$

$$\frac{|n-1|}{n-1} = \frac{n-1}{n-1} = 1$$

$n \rightarrow 1^+ \rightarrow 1$
 $n \rightarrow 1^- \rightarrow -1$

$$\frac{|n-1|}{n-1} = \frac{-(n-1)}{n-1} = -1$$

$n \rightarrow 1^+ \rightarrow 1$
 $n \rightarrow 1^- \rightarrow -1$

e quindi

$$\lim_{x \rightarrow 1^+} \frac{1 - \cos \sqrt{x-1}}{e^{x-1} - 1} =$$

$$= \lim_{x \rightarrow 1^+} \frac{1 - \cos \sqrt{x-1}}{(\sqrt{x-1})^2} \cdot \frac{1}{e^{x-1} - 1} \cdot \frac{|x-1|}{x-1}$$

$$= \frac{1}{2} \cdot \frac{1}{1} \cdot 1 = \frac{1}{2}$$

$$\lim_{n \rightarrow 1^-} \frac{1 - \cos \sqrt{n-1}}{(\sqrt{n-1})^2} \cdot \frac{1}{\frac{e^{n-1} - 1}{n-1}} \cdot \frac{|n-1|}{n-1} =$$

$$= \frac{1}{2} \cdot \frac{1}{1} \cdot (-1) = -\frac{1}{2}$$

però

~~$\lim_{n \rightarrow 1}$~~

$$\frac{1 - \cos(\sqrt{n-1})}{e^{n-1} - 1}$$

Ex

Ω_n

$\frac{2}{\pi}(3\pi)$

$\int 0 \int$

$$\lim_{x \rightarrow 0} \frac{x}{1 - \cos x}$$

(0/0)

$$\frac{t_f^2(3x)}{1 - \cos x} = \frac{t_f^2(3x)}{\frac{1 - \cos x}{x^2} \cdot x^2} =$$

$$= \frac{1}{1 - \cos x} \cdot \frac{\sin^2(3x)}{x^2} =$$

$t_f^2 3x = \frac{\sin^2}{\cos^2} (3x)$
~~cos~~

$$= \frac{1}{1 - \cos x} \cdot \frac{1}{\cos^2(3x)} \cdot \frac{\sin^2(3x)}{x^2}$$

$$= \frac{1}{1 - \frac{\cos n}{n^2}} \cdot \frac{1}{\cos^2(3n)} \cdot \frac{\sec^2(3n)}{(3n)^2} \cdot \frac{(3n)^2}{n^2} \rightarrow \frac{1}{1^2} \cdot 1^2 \cdot 9 = 9$$

Diagram illustrating the limit calculation using the product rule for limits. The expression is broken down into four factors, each enclosed in a circle. Arrows indicate the limit of each factor as $n \rightarrow \infty$:

- Factor 1: $\frac{1}{1 - \frac{\cos n}{n^2}} \rightarrow \frac{1}{1^2}$
- Factor 2: $\frac{1}{\cos^2(3n)} \rightarrow 1^2$
- Factor 3: $\frac{\sec^2(3n)}{(3n)^2} \rightarrow 1^2$
- Factor 4: $\frac{(3n)^2}{n^2} \rightarrow 9$

Per tanto

$$\lim_{n \rightarrow \infty} \frac{\tan^2(3n)}{1 - \cos n} = 2 \cdot 1 \cdot 1 \cdot 9 = 18$$

T (caratterizzazione sequenziale dei limiti)

Se $f: X \rightarrow \mathbb{R}$, $x_0^* \in \mathbb{R}^*$ ($x_0^* = x_0 \in \Delta X$)

$\exists \lim_{x \rightarrow x_0^*} f(x) = l^* \iff$
 "x è solo x"
 oppure
 "C.N.S."

Per ogni successione $\{x_n\} \subseteq X$ tale che

$(x_n \neq x_0 \quad \forall n \in \mathbb{N})$

$x_n \rightarrow x_0^*$

allora

$\exists \lim_{n \rightarrow \infty} f(x_n) = l^*$

U2>

Se $\exists \{a_n\}, \{b_n\} :$

$$a_n \rightarrow n_0^*$$

$$b_n \rightarrow n_0^*$$

Se $f(a_n) \rightarrow l_1^*$

$f(b_n) \rightarrow l_2^*$

con $l_1^* \neq l_2^*$

allora ~~$\nexists$~~ $\lim_{n \rightarrow n_0^*} f(n)$

Ex

$$\lim_{n \rightarrow 0} \sin\left(\frac{1}{n}\right)$$

Provare che non esiste.

$$a_n = \frac{1}{\frac{\pi}{2} + 2n\pi} \xrightarrow{n \rightarrow \infty} 0$$

$$b_n = \frac{1}{2n\pi} \xrightarrow{n \rightarrow \infty} 0$$

$$f(a_n) = \operatorname{Re}\left(\frac{1}{\frac{1}{\frac{\pi}{2} + 2n\pi}}\right) = \operatorname{Re}\left(\frac{\pi}{2} + 2n\pi\right) = \operatorname{Re}\left(\frac{\pi}{2}\right) = 1$$

$$\lim_{n \rightarrow \infty} f(a_n) = \lim_{n \rightarrow \infty} \operatorname{Re}\left(\frac{\pi}{2} + 2n\pi\right) = 1$$

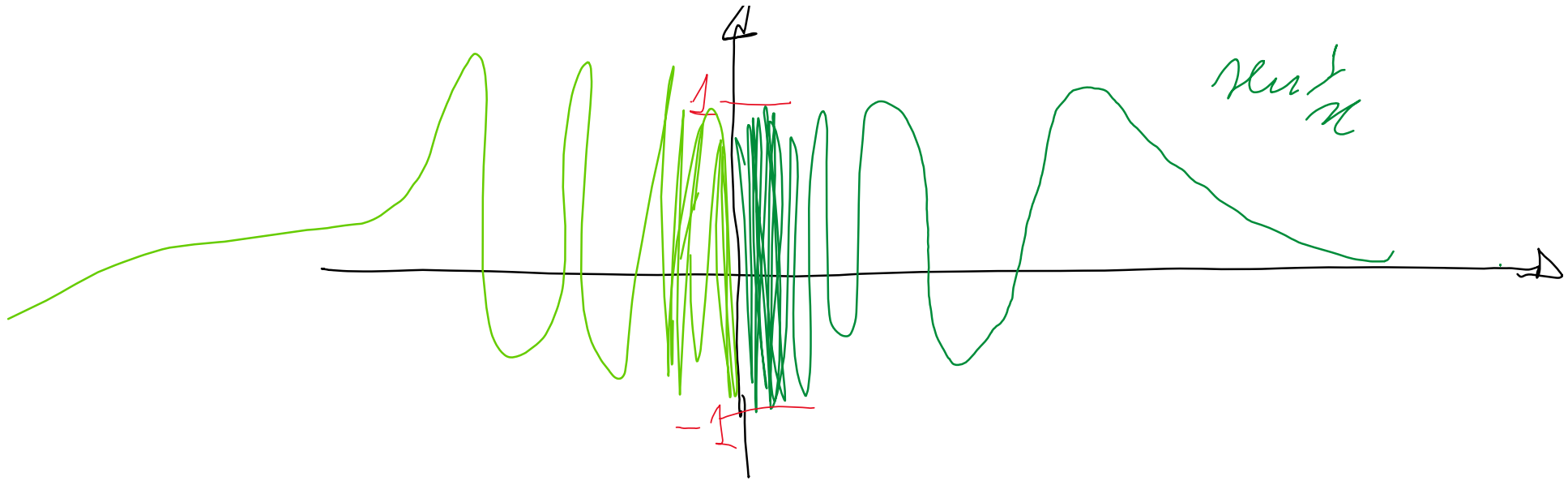
$$f(b_n) = \sin\left(\frac{1}{\frac{1}{2n\pi}}\right) = \sin(2n\pi) = \sin(0) = 0$$

$$\exists \lim_{n \rightarrow \infty} f(b_n) = \lim_{n \rightarrow \infty} \sin(2n\pi) = 0$$

Però

$$1 = \lim_{n \rightarrow \infty} f(a_n) \neq \lim_{n \rightarrow \infty} f(b_n) = 0$$

$$\Rightarrow \not\lim_{n \rightarrow 0} \sin\left(\frac{1}{x}\right)$$



Il precedente teorema permette di avere

(CARATTERIZZAZIONE ~~spaziabile~~
della CONTINUITÀ)

Se $f: X \rightarrow \mathbb{R}$, $x_0 \in X$, $x_0 \in \mathcal{D}X$

f è continua in $x_0 \iff$

funzione continua in x_0

Per ogni $\{x_n\} \subseteq X$ tale che $x_n \rightarrow x_0$

$$\exists \lim_{n \rightarrow \infty} x_n = x_0$$

allora $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$

DEF

Se $f: X \rightarrow \mathbb{R}$, $x_0 \in \Delta X$

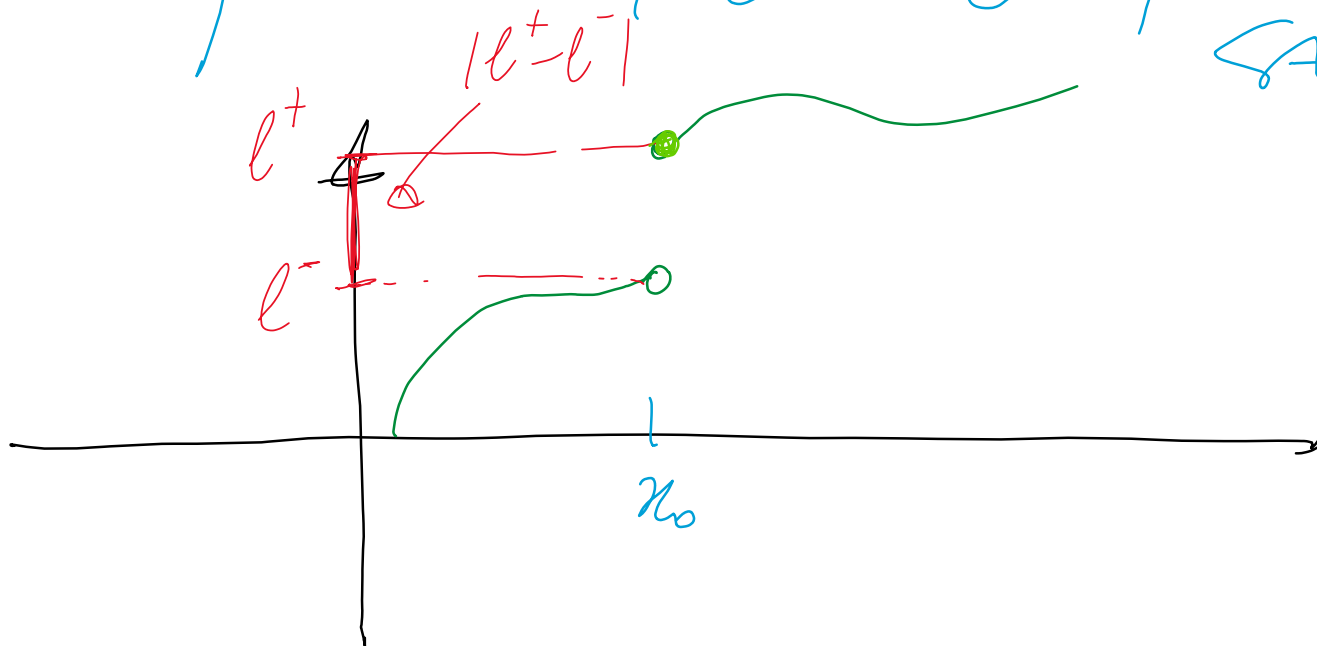
(a) Se $\exists \lim_{x \rightarrow x_0^+} f(x) = l^+ (\in \mathbb{R})$

$\exists \lim_{x \rightarrow x_0^-} f(x) = l^- (\in \mathbb{R})$

non $l^+ + l^-$ (se non esiste)

con $x \rightarrow c$ ($A \rightarrow \infty$)

Dimostrare che f ha una DISCONTINUITÀ
DI I SPECIE (O DI SALTO)
e la quantità $|l^+ - l^-|$ si dice
SALTO



b) $\exists \lim_{x \rightarrow c} f(x) = \pm \infty$

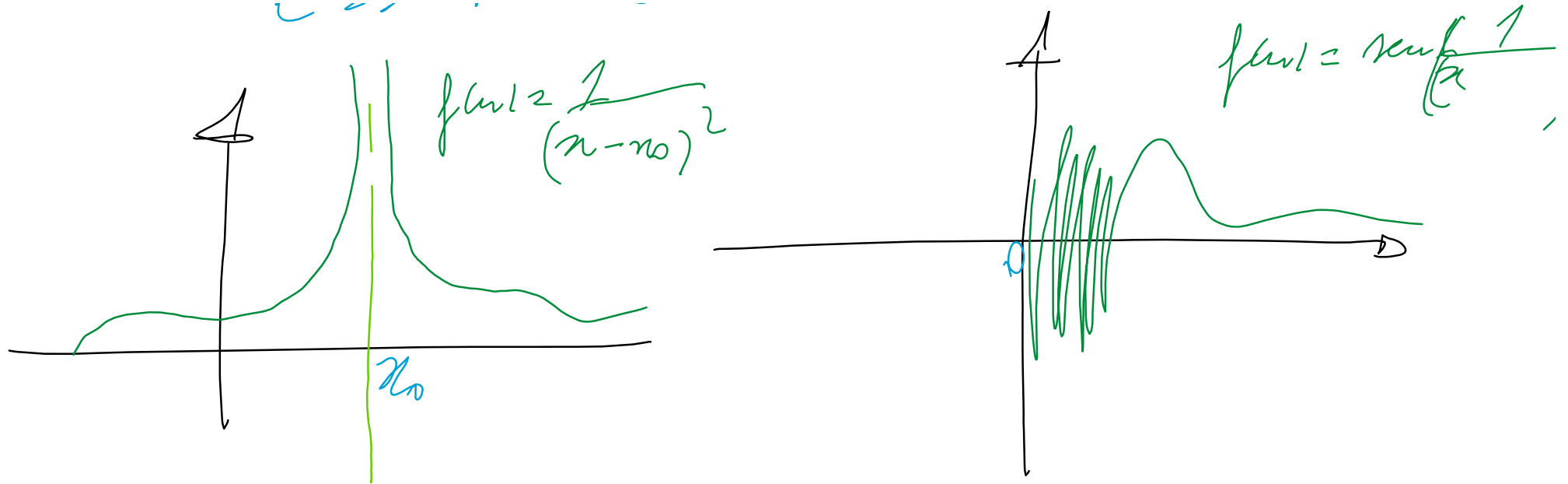
$$\left(\exists \lim_{n \rightarrow n_0^+} f(n) = \pm \infty \right)$$

allora f HA UNA DISCONTINUITA'
DI II SPECIE INFINITA

Ma anche $\nexists \lim_{n \rightarrow n_0^-} f(n)$

$$\left(\nexists \lim_{n \rightarrow n_0^+} f(n) \right)$$

f HA una DISCONTINUITA' DI II SPECIE
ESSENZIALE

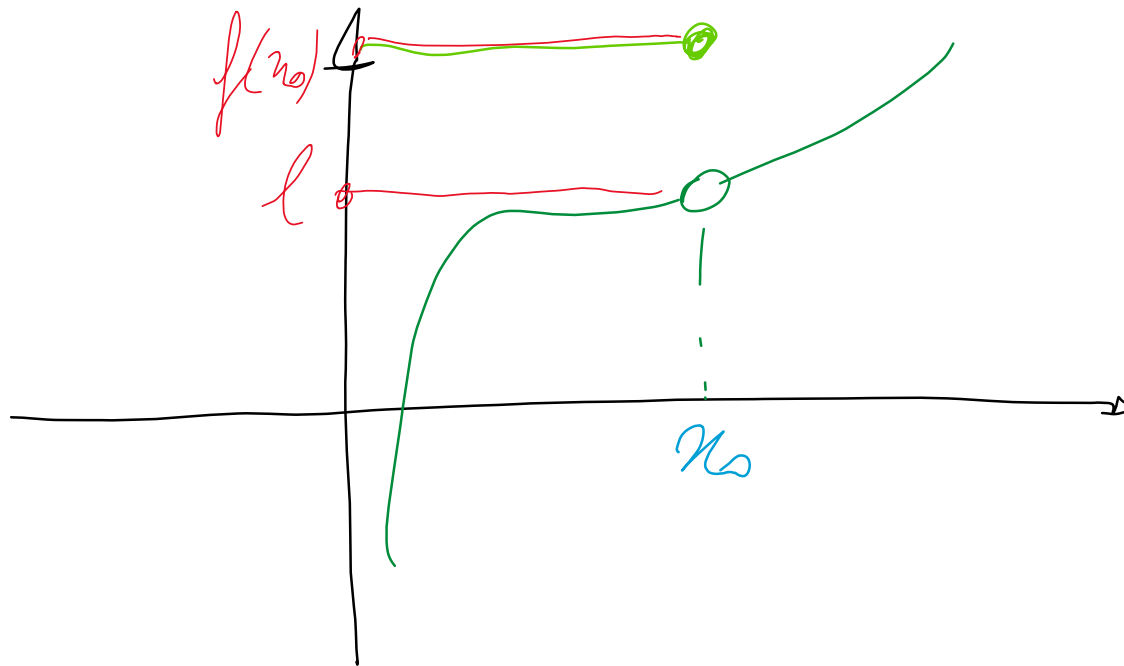


c) $\exists \lim_{x \rightarrow x_0} f(x) = l \quad (l \in \mathbb{R})$

me $f(x_0) \neq l$

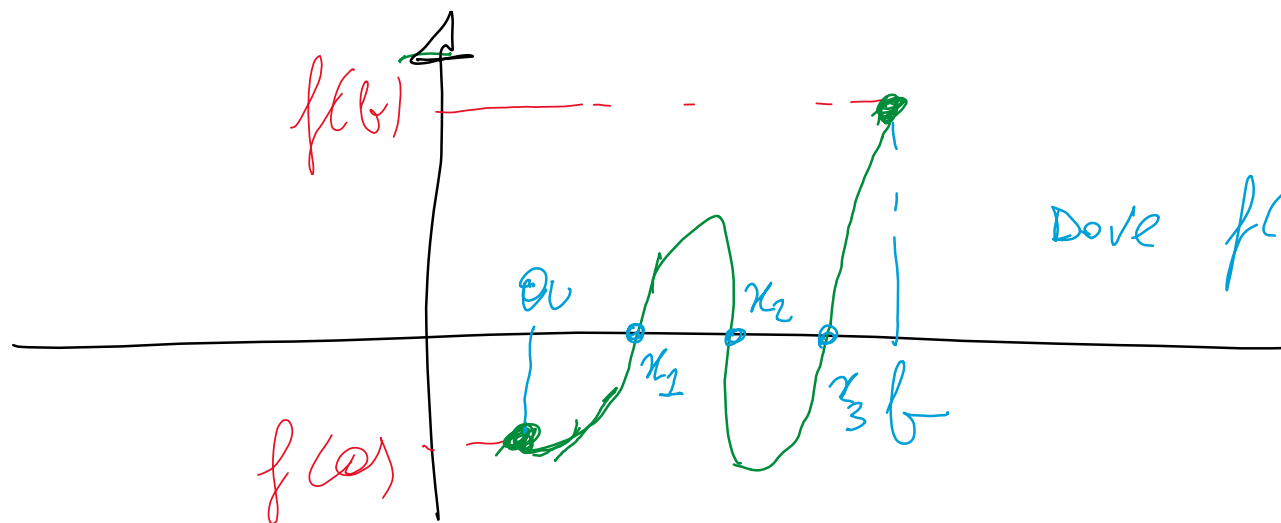
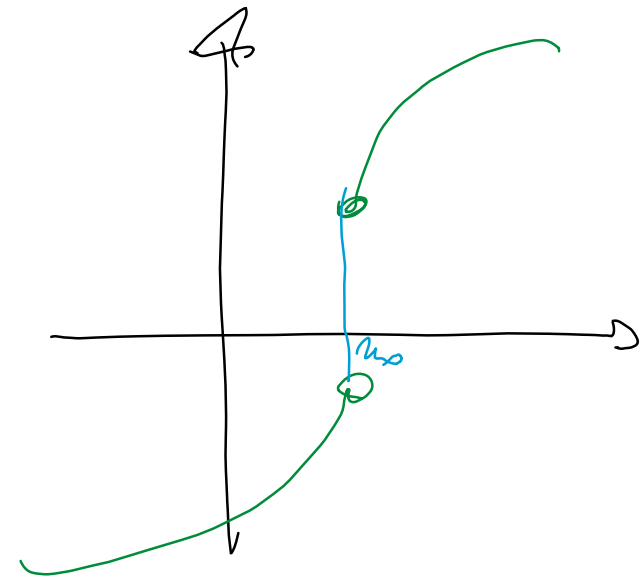
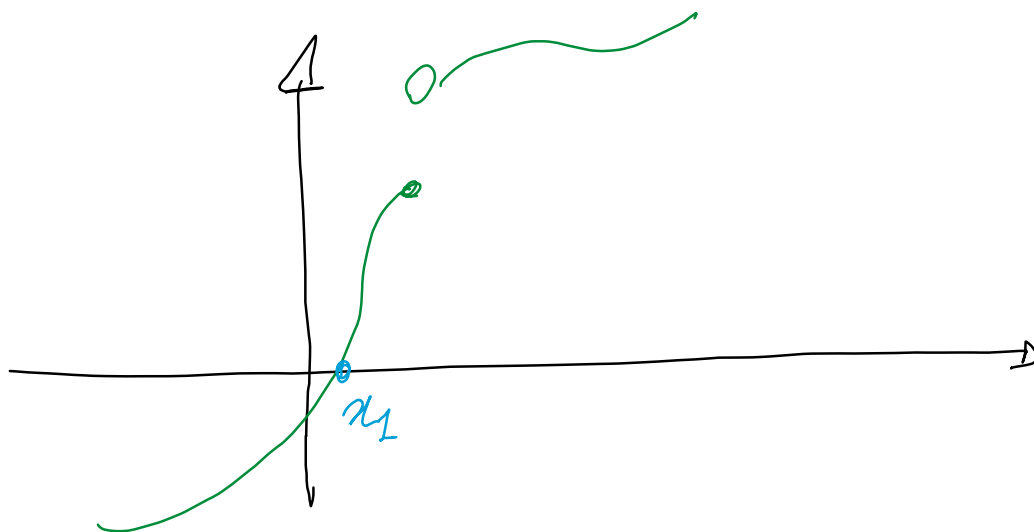
oppure f non è definita in x_0

allora f HA UNA DISCONTINUITA'
DI III SPECIE ELIMINABILE



PROPRIETA' GLOBALI
DELLE FUNZIONI CONTINUE

(SO DEFINITE IN INTERVALLI)



Dove $f(x_1) = f(x_2) = f(x_3) = c$

T (di continue degli zeri)

Se $f: (a, b) \rightarrow \mathbb{R}$
 f CONTINUA IN (a, b)

$\exists a, b \in (a, b)$ tali che
 $f(a) \cdot f(b) < 0$, allora

$\exists x_0 \in]a, b[: f(x_0) = 0$

DEF x_0 INDICE ZERO PER f
 (ovvero x_0 è una soluzione
 dell'equazione $f(x) = 0$)

DIJ (per "dicotomia"
 o "BISSEZIONE")

Indichiamo con $a_0 = a$

$b_0 = b$

e per semplicità $f(a) < 0$

$f(b) > 0$

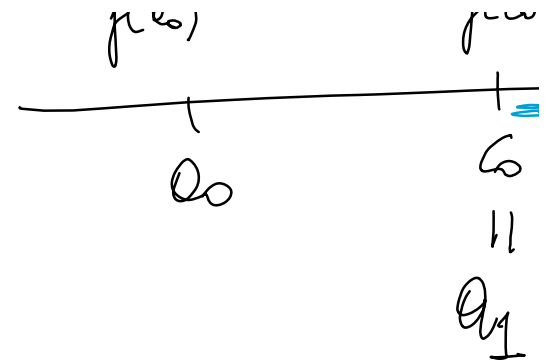
(6) Calcoliamo
~~la~~

$$c_0 = \frac{a_0 + b_0}{2}$$

$f(a) < 0$



$$\Rightarrow f(c) < 0$$



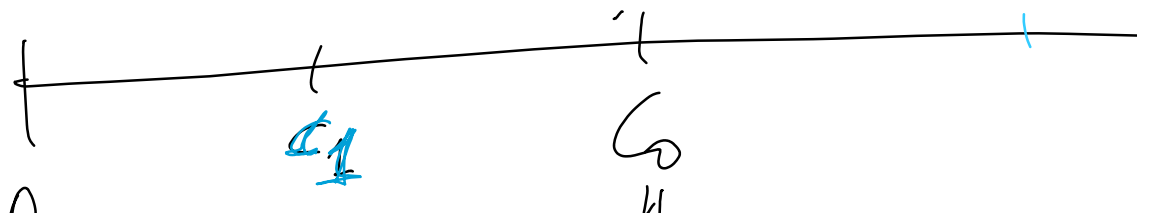
$$c_0 < c_1 < b_1 = b_0$$

$$f(c_1) f(b_1) < 0$$

$$b_1 - c_1 = \frac{1}{2} (b_0 - c_0)$$

Calcolo

$$c_1 = \frac{b_1 + c_1}{2}$$



$$\begin{array}{ccc} \begin{array}{c} \cup \\ \cap \\ e_1 \end{array} & \begin{array}{c} \cap \\ a_2 \end{array} & \ddot{b}_1 \\ \text{If } f(c_1) = 0 & & \text{STOP} \end{array}$$

$$\begin{array}{ccc} \text{If } f(c_1) > 0 & & a_2 = a_1 \\ & & b_2 = c_1 \end{array}$$

$$a_0 \leq a_1 \leq c_2 < b_2$$

$$f(c_2) f(b_2) < 0$$

Indipendente

$$f(c_2) < 0$$

In generale, si pone n

$$n) \quad a_{n+1} < b_{n+1}$$

$$(0) \quad f(a_{n+1}) < 0, \quad f(b_n)$$

$$(1) \quad a_0 \leq a_1 \leq \dots \leq a_{n+1} < b_{n+1}$$

$$(2) \quad b_{n+1} - a_{n+1} = \frac{1}{2}(b_n - a_n)$$

$$C_{n+1} = \frac{a_n + b_n}{2}$$

è continuando a prendere il c
 di due successioni $\{a_n\}$ e
 che hanno queste caratteristiche

$\{a_n\}$ è crescente

$\{b_n\}$ è decrescente con

$$a_0 \leq a_1 < \dots < a_n < b_n \leq b_{n-1}$$

$$a_0 \leq a_n \leq b_0$$

$$b_0 \leq b_n \leq b_0$$

per questo possiamo concludere che

$$\exists \lim_{n \rightarrow +\infty} a_n = \sup a_n = x_1$$

$a_n \in \mathbb{R}$
monotona

$$\exists \lim_{n \rightarrow +\infty} b_n = \inf b_n = x_2 \in \mathbb{R}$$

Provare che $x_0 = x_1$. Infatti

$$b_n - c_n \stackrel{!}{=} \frac{1}{2} (b_{n-1} - c_{n-1}) \stackrel{!}{=} \frac{1}{2} \left(\frac{1}{2} (b_{n-2} - c_{n-2}) \right)$$

$$b_n - c_n = \left(\frac{1}{2} \right)^n (b_0 - c_0)$$

Pertanto

$$\begin{array}{ccccc}
 b_n & = & \left(\frac{1}{2} \right)^n & (b_0 - c_0) & + & c_n \\
 \downarrow & & & \downarrow & & \downarrow \\
 x_1 & & & 0 & & x_0
 \end{array}$$

$\Rightarrow$ ha l'incisione del limite delle

$$x_1 = x_0$$

$$\left(\begin{array}{l} a_n \rightarrow x_0 \\ b_n \rightarrow x_0 \end{array} \right)$$

Provare, infine, che x_0 è lo zero

$$f(a_n) < 0 \quad \forall n \in \mathbb{N}$$

$$f(b_n) > 0$$

$$f(x_n) \rightarrow f(x_0)$$

per il T. di Cauchy
esprimibile delle conti

$$f(x_n) \rightarrow f(x_1) = f(x_0)$$

è quindi

$$f(x_n) f(x_m) < \epsilon$$

$$f(x_0) \quad f(x_0)$$

$$f(l_n) f(b_n) \rightarrow [f(x_0)]^2 =$$

T (PERTINENZA DEL SEGNO)

$$\& f(x) < K \quad \forall x \in \mathcal{D}_{f(x)}$$

$$\& \exists \lim_{n \rightarrow \infty} f(x) = l$$

$$l \leq K$$

$$M_n \cap \mathcal{D}_f \subset \mathcal{D}_f$$

$$\forall \forall \quad \lfloor f(x_0) \rfloor = 0 \iff f(x_0)$$

Ex

$$(1) \quad x^3 - 2x = \log_2 x \quad \forall$$

$$f(x) = x^3 - 2x - \log_2 x \in]0, +\infty[$$

$\bar{e}$ CONTINUA

ogni zero di f $\bar{e}$ UNA SOLA

$$f(1) = 1^2 - \log_2(1) = -1 <$$

$$f(2) = 2^2 - 4 - \log_2(2) = 4 - 1 =$$

$$a = 1, \quad b = 2 \quad \text{dove}$$

$$f(1) f(2) < 0$$

per il T.E.Z.

$$\exists x_0 \in]1, 2[= f(x_0)$$

$$f\left(\frac{3}{2}\right) = \frac{27}{8} - 3 - \log_2\left(\frac{3}{2}\right)$$

$$= \frac{27}{8} - 3 - (\log_2 3 -$$

$$= \frac{27}{8} - 3 - \log_2 3 + 1$$

$$x_0 \in \left] \frac{3}{2}, 2 \right]$$

$$\frac{3}{2} + 1$$

Del T-E-L. discente

T(dei valori INTERMEDI O PER
DARI.

1 1 / . n 1 1 1

Se $f: (a, b) \rightarrow \mathbb{R}$, CONTINUA

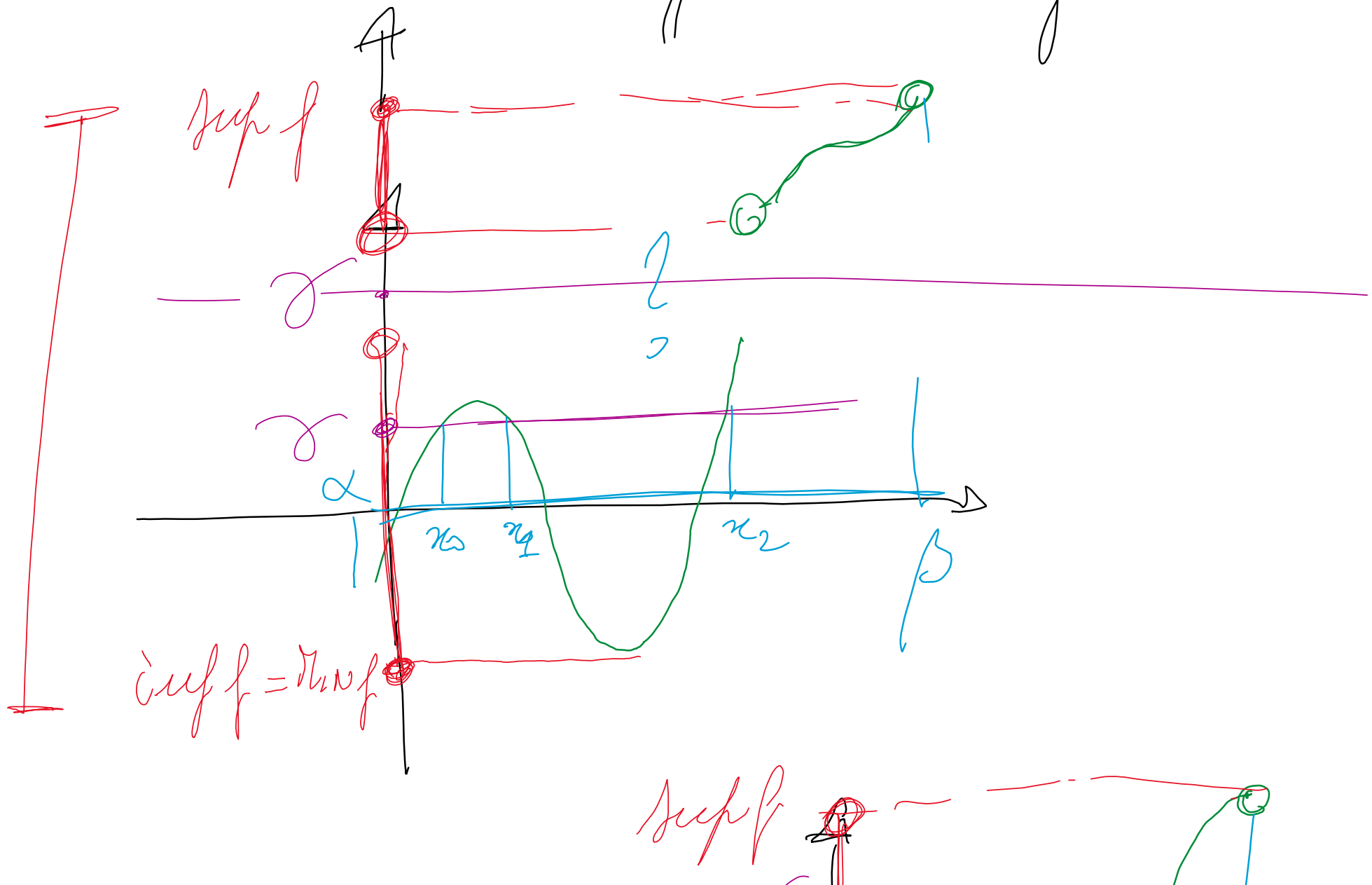
allora f ASSUME TUTTI I

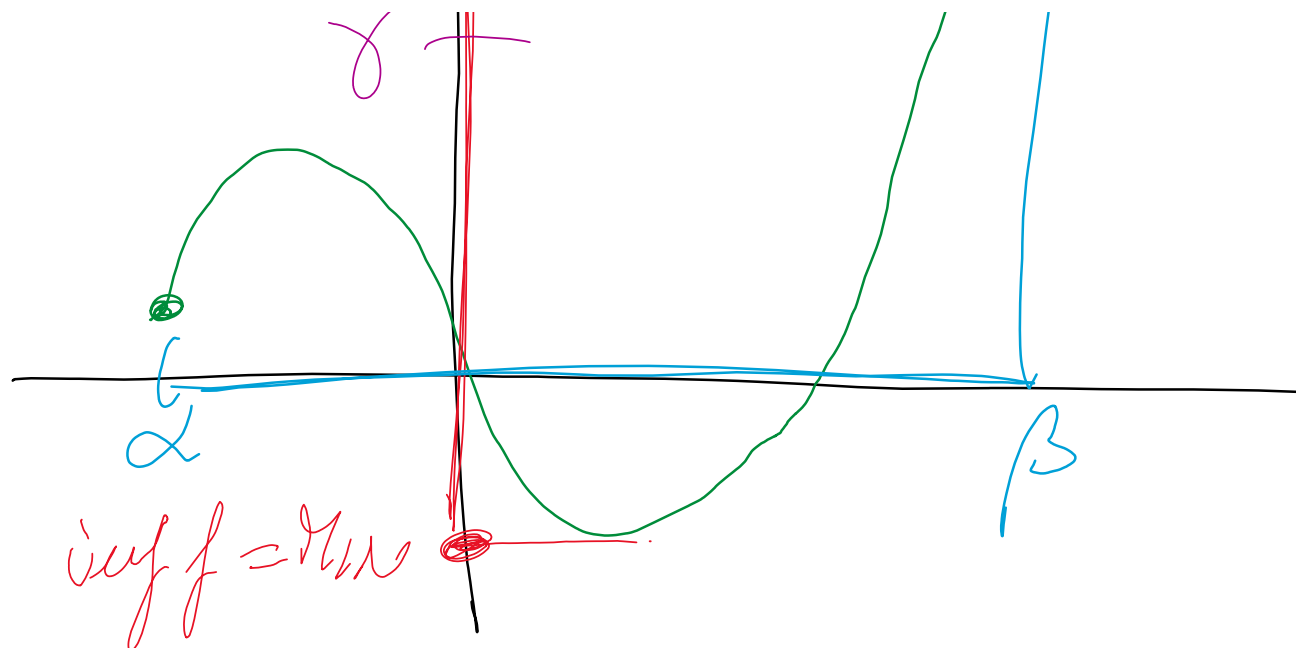
COMPRESSI TRA $\inf f$ e $\sup f$ in (a, b)

ovvero

$\forall \gamma \in (\inf f, \sup f)$ in (a, b)

$$\exists x_0 \in (a, b) : f(x_0) = \sup f$$





D, V

Le $\gamma \in (\inf_{\alpha, \beta} f, \sup_{\alpha, \beta} f)$

allora (PROVARE SE CI SI PUO' S.
USARE PROP DEI SUP E INF)

$$\exists y_1, y_2 \in \text{Im}(f)$$

~~tale~~ tale che $y_1 < \gamma <$

$$\text{Se } h(x) = f(x) - \gamma \in (a, \beta)$$

Per farlo che $y_1, y_2 \in \text{Im}(f) \Rightarrow$

$$\exists x_1 \in (a, b) : f(x_1) = y_1$$

$$\exists x_2 \in (a, b) : f(x_2) = y$$

allora h soddisfa le ipotesi del

$h(x)$ è CONTINUA in $(a,$

$$h(x_1) = f(x_1) - \gamma = y_1 - \gamma =$$

$\wedge$

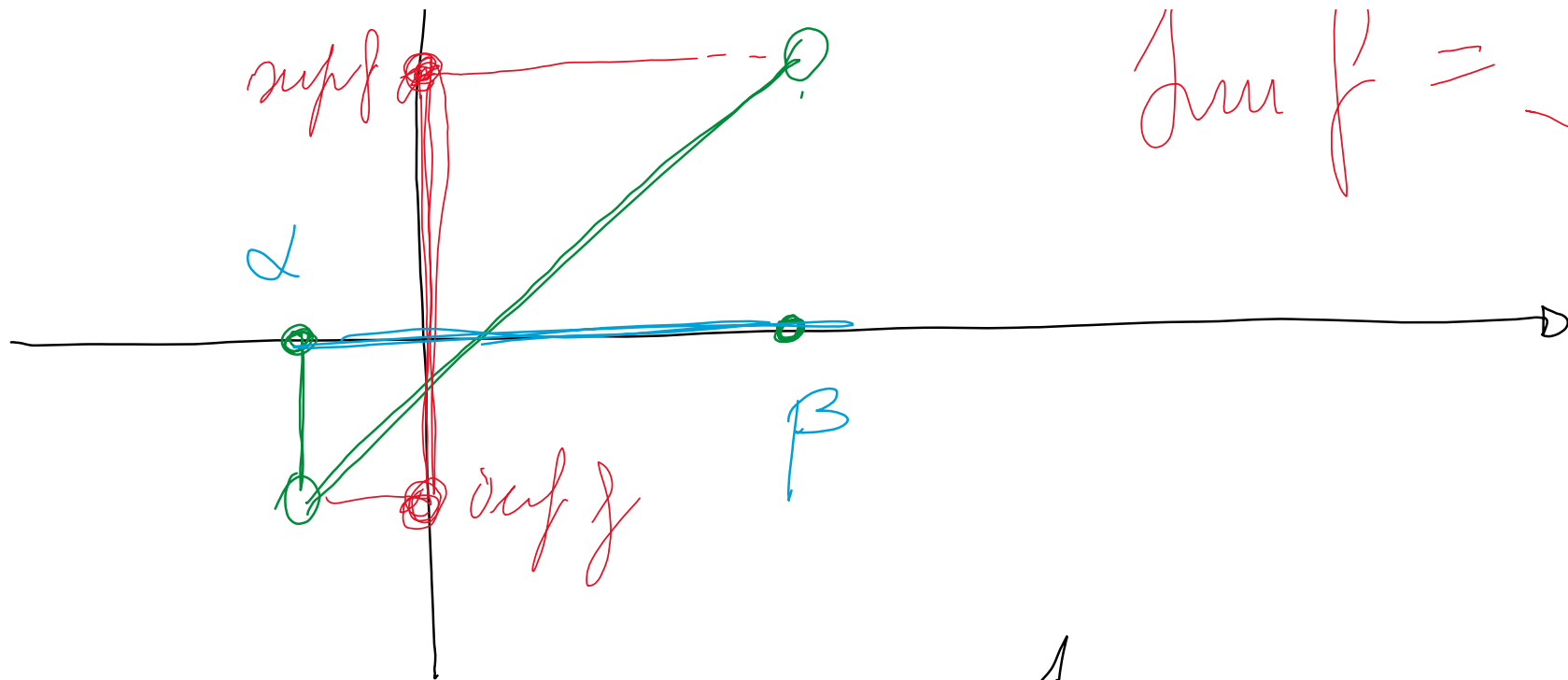
$$h(x_2) = f(x_2) - \gamma = y_2 - \gamma =$$

quindi $\exists x_0 \in]x_1, x_2[:$

$$h(x_0) = 0$$

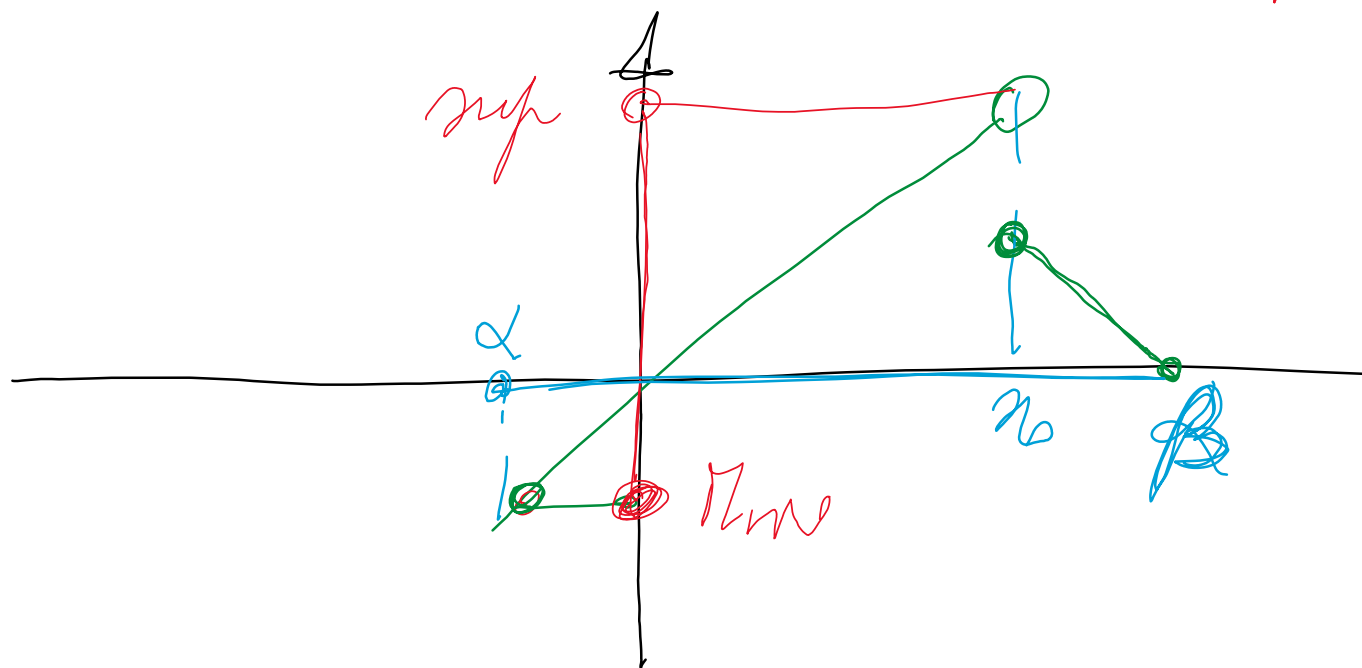
$$0 = h(x_0) = f(x_0) - \gamma \Rightarrow f(x_0) = \gamma$$

4

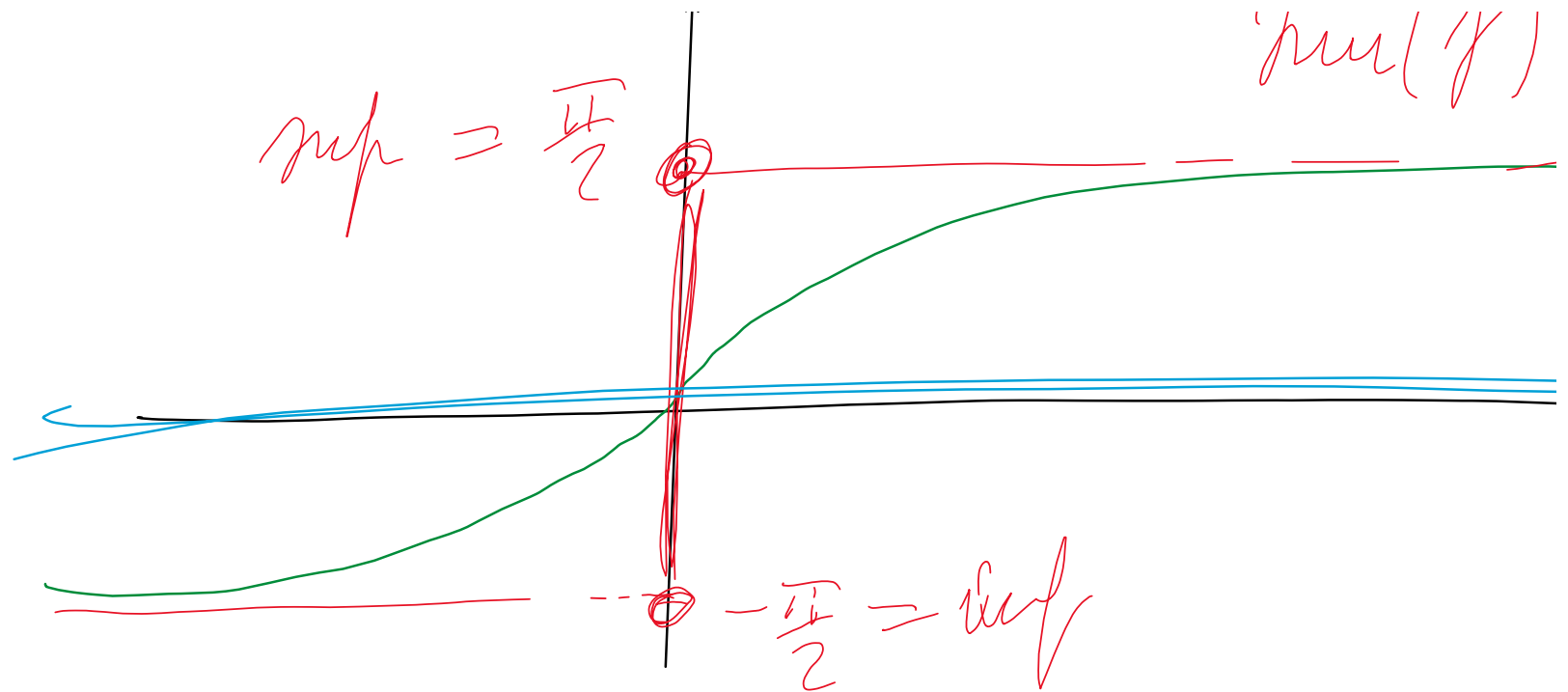


$$\int f = \int \inf$$

$\mathbb{R}$



$\mathbb{R}$ / 1.1



T (dw Weierstrass)

Se $f: [a, b] \rightarrow \mathbb{R}$ $[a, b]$

f CONTINUA IN $[a, b]$,

$$\text{Im}(f) = [\min_{[a, b]} f, \max_{[a, b]} f]$$

Allo

f ammette Massimo $\in \mathbb{R}$

Cvè

$$\exists x_0 \in [a, b] : f(x_0) \geq f(x)$$

$$\neg \dots \neg \dots \neg \dots \neg \dots \neg \dots \neg \dots \neg \dots \neg \dots \neg \dots$$

$$\exists n_1 \in \mathbb{N}, v_j : f(n_1) \leq f$$

Def

(OMSSA)

Let $f: (a, b) \rightarrow \mathbb{R}$, $x_0 \in]a$

DEF

1

1

1

1

1

1

1

1

1

1

1

$$R_f(x, x_0) = \frac{f(x) - f(x_0)}{x - x_0} := Q$$
$$f(x) = x^3 : \mathbb{R} \rightarrow \mathbb{R} \quad x_0 = 5$$

$$R_f(5, x) = \frac{f(x) - f(5)}{x - 5} =$$

$$= \frac{\quad}{\quad}$$

$$f(x) = \sin x \quad x_0 = \pi$$

$$R_f(\pi, x) = \frac{f(x) - f(\pi)}{x - \pi} = \frac{\sin x}{\quad}$$

f

 $x - n$

$$= \frac{1}{n-}$$

Un modo equivalente di scrivere la
FUNZIONE RAPPORTO INCREMENTALE

è il seguente

$$f(n_0 + b) - f(n_0)$$

$$K_f(n_0, h) = \frac{0}{h} \quad \quad \quad : (a - n_c)$$

OVVIA MENTE BASTA CONSIDERAR

$$h = x - n_0 \Leftrightarrow x =$$

PER PASSARE DA UNA ESPRESSIONE

Ex

$$|x - n| = a \vee n$$

$$n_0 = \overline{n}$$

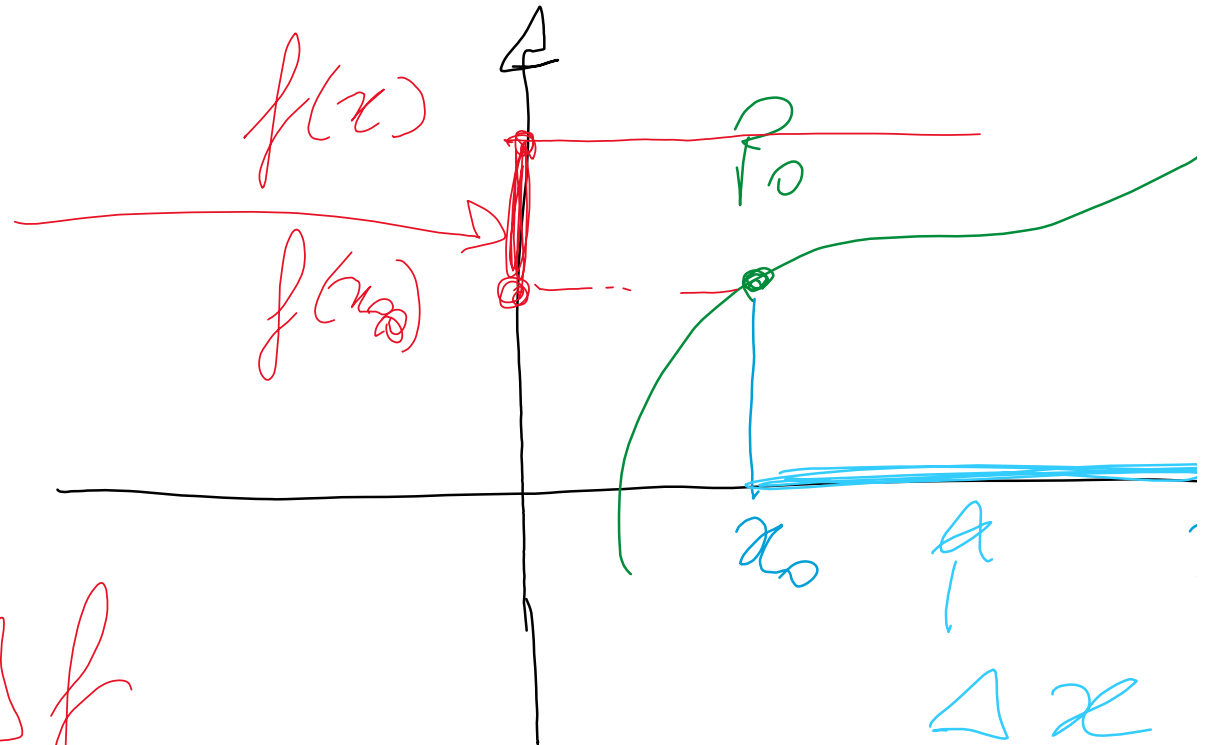
$$\begin{aligned}
 R_f(\pi, h) &= \frac{f(\pi+h) - f(\pi)}{h} = \\
 &= \frac{\sin(\pi+h) - \sin \pi}{h} =
 \end{aligned}$$

OSS

Il rapporto incrementale π
 UN TASSO DI VARIAZIONE

una VELOCITA' DI VARIAZIONE

$$\Delta f = f(x) - f(x_0)$$



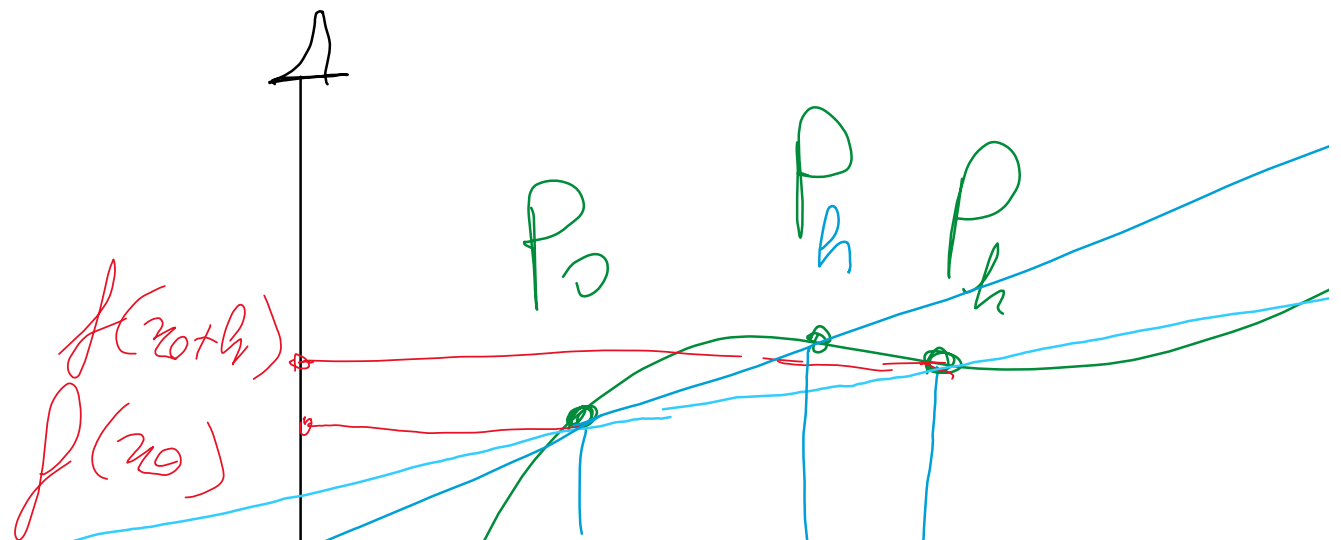
$$R_f(x_0, x) = \frac{\Delta f}{\Delta x}$$

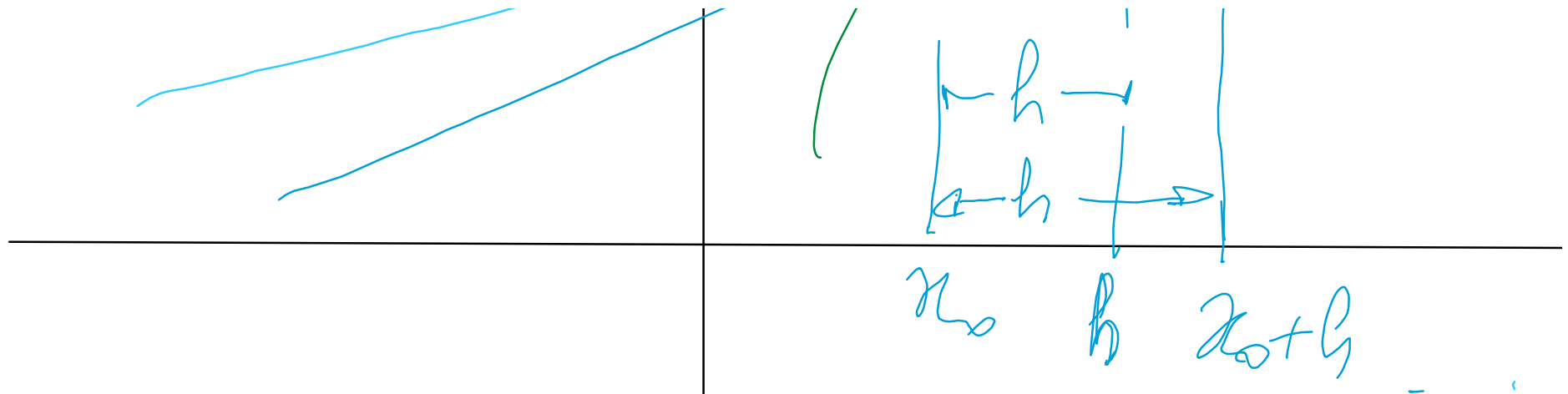
Attenzione:

OSS

$$\tilde{R}_f(x_0, u) = R_f(x_0, h)$$

Chw $\bar{e}$ $R_f(x_0, h)$? per il f
 me





A, B

$$| \quad \Delta_{AB} = y - y_A = \frac{y_B - y_A}{x_B - x_A} (x$$

$$\left(\frac{y - y_A}{x - x_A} = \frac{y_B - y_A}{x_B - x_A} \right)$$

$$A = P_0$$

$$B = P_h$$

$$\sum_{P_0 P_h}$$

$$y - f(x_0)$$

$$= \frac{f(x_0+h) - f(x_0)}{h}$$

$$\sum_{P_0 P_h} = y - f(x_0) = \frac{f(x_0+h) - f(x_0)}{h}$$

è l'equazione delle rette Sec
grafico di f che pone per P_0

Da questi esempi grafici mostra

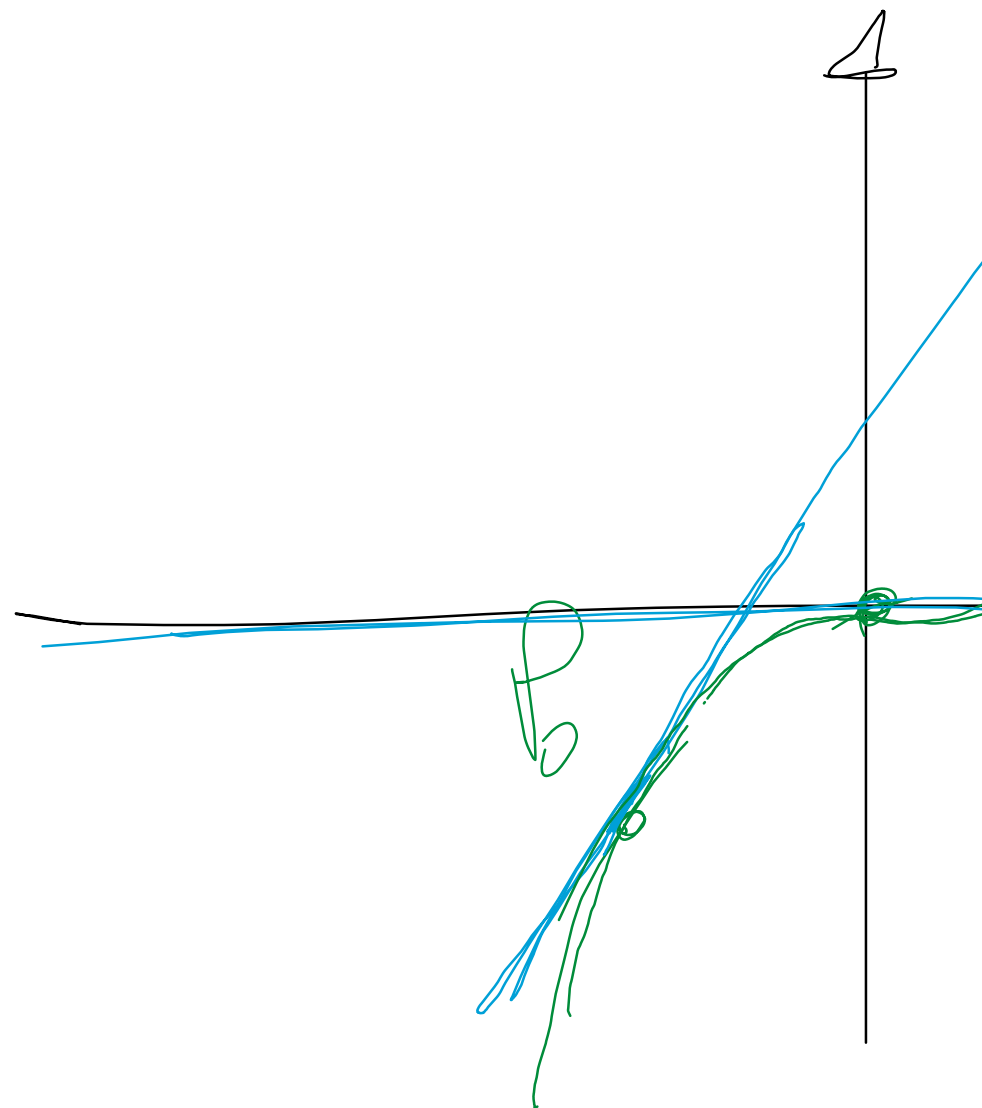
che

$R_f(x, h)$

RAPPRESENTA

il Coefficiente ANG

di una retta secante



Per le parabole cubiche,
incontrate sempre
il grafico in,